

LECK AP, Germany

WMO: 100220

Lat: 54.7903N

Lon: 8.9514E

Elev: 26

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	15.3	19.6	11.9	10.1	19.1	15.8	12.2	21.2	33.3	39.2	30.6	40.7	5.1	90	0.649

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.0	80.7	66.7	76.9	65.0	73.1	63.4	68.3	77.4	66.4	74.0	64.7	71.0	10.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.1	93.0	71.6	63.5	87.8	69.6	61.9	83.0	67.9	32.5	77.2	31.1	74.2	29.7	71.0	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.6	24.6	21.9	8.3	85.9	5.9	2.9	4.0	87.9	0.6	89.6	-2.8	91.2	-7.1	93.3
			8.0	72.2	5.9	1.5	3.7	73.3	0.3	74.1	-3.0	75.0	-7.3	76.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.5	34.4	34.5	38.3	45.7	52.2	57.2	62.2	62.3	56.6	49.0	41.7	35.1
	DBStd	11.84	7.24	7.21	5.95	6.04	5.31	5.06	4.91	4.87	4.50	5.85	6.14	7.64
	HDD50	2286	484	435	364	157	37	2	0	0	4	88	254	461
	HDD65	6459	949	855	828	579	399	241	118	116	253	495	700	926
	CDD50	1375	0	0	1	28	104	218	379	380	203	58	4	0
	CDD65	74	0	0	0	0	1	8	32	31	2	0	0	0
	CDH74	649	0	0	0	2	13	85	296	243	10	0	0	0
	CDH80	115	0	0	0	0	0	10	59	46	0	0	0	0
Wind	WSAvg	10.1	11.4	12.0	12.0	10.4	9.8	9.5	8.6	8.3	9.0	9.9	10.2	10.2
Precipitation	PrecAvg	33.5	2.8	2.2	1.9	1.5	2.0	2.7	3.1	3.9	3.5	3.7	3.2	3.2
	PrecMax	38.9	5.6	4.1	3.8	3.9	3.5	4.4	5.5	8.0	7.4	7.3	6.7	6.8
	PrecMin	21.0	0.1	0.6	0.4	0.2	0.3	0.0	1.1	1.0	0.6	1.4	0.6	1.1
	PrecStd	4.4	1.3	0.9	1.0	0.8	0.8	1.2	1.4	1.8	1.7	1.3	1.5	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.2	49.0	57.6	70.9	75.7	81.4	85.8	84.6	75.2	64.9	56.8	51.3
		MCWB	47.0	46.7	48.9	57.1	61.7	66.4	67.9	67.6	64.3	60.7	54.7	49.9
	2%	DB	46.9	46.8	52.3	65.2	70.0	76.1	80.9	80.6	70.8	61.3	53.7	48.7
		MCWB	45.4	45.2	47.4	53.5	58.7	64.2	66.8	66.9	62.4	58.0	51.7	47.3
	5%	DB	45.2	45.3	48.5	60.4	66.4	71.1	77.2	76.0	67.0	59.1	51.8	46.9
		MCWB	44.0	43.8	45.0	52.1	55.9	61.7	65.3	65.4	60.7	56.7	50.1	45.6
	10%	DB	43.7	43.7	46.2	56.0	62.9	67.1	72.9	71.9	64.3	57.3	50.1	45.1
		MCWB	42.5	42.4	43.8	50.0	54.7	59.8	63.7	64.1	59.2	54.9	48.4	43.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.9	47.5	51.3	58.7	63.7	69.2	70.4	70.6	66.2	61.2	54.9	50.2
		MCDB	48.9	48.5	55.2	68.1	71.9	78.1	80.3	79.5	72.3	63.7	56.1	51.2
	2%	WB	45.7	45.5	48.0	55.4	60.1	65.8	68.4	68.0	63.5	58.9	52.2	47.6
		MCDB	46.5	46.4	51.2	62.4	66.8	72.7	78.2	76.7	68.5	60.4	53.3	48.5
	5%	WB	44.2	44.2	45.8	52.9	57.6	62.7	66.6	66.3	61.8	57.0	50.3	45.7
		MCDB	45.0	45.1	48.1	58.8	63.8	69.1	74.7	73.6	66.0	58.6	51.4	46.6
	10%	WB	42.7	42.5	44.2	50.5	55.7	60.5	64.5	64.7	60.1	55.4	48.7	44.1
		MCDB	43.6	43.5	45.9	55.2	61.4	66.0	71.0	70.8	63.7	57.0	49.8	45.0
Mean Daily Temperature Range	5% DB	MDBR	7.4	8.2	10.8	15.0	15.5	14.1	15.0	15.2	13.1	11.2	8.5	8.2
		MCDBR	6.6	7.3	15.0	22.1	22.5	22.4	23.8	22.1	17.5	11.7	8.2	7.0
		MCWBR	6.4	6.3	10.6	13.0	13.2	12.4	11.6	11.0	10.6	9.2	7.6	7.0
	5% WB	MCDBR	6.4	6.9	12.8	19.2	18.5	19.1	20.8	19.1	15.0	10.7	7.7	7.2
		MCWBR	6.4	6.3	9.8	12.2	12.2	12.1	11.1	10.5	9.8	9.0	7.5	7.2
Clear-Sky Solar Irradiance	taub	0.325	0.344	0.369	0.402	0.393	0.391	0.406	0.405	0.384	0.361	0.338	0.315	
	taud	2.269	2.301	2.273	2.235	2.321	2.350	2.323	2.336	2.377	2.384	2.322	2.268	
	Ebn at Noon	189	231	252	260	270	272	264	257	246	222	183	166	
	Edh at Noon	19	25	32	39	37	37	37	35	29	23	18	15	
All-Sky Solar Radiation	RadAvg	161	361	773	1292	1652	1722	1616	1307	923	491	200	117	
	RadStd	22	52	69	146	163	141	184	99	100	57	26	10	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-263	N/A	N/A

Nomenclature: See separate page